Variational Autoencoders and Nonlinear ICA: Identifiability in Deep Latent Variable Models

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Motivation: Why Study Latent Variables?

Many real-world datasets are generated from latent factors:

- Images may vary due to lighting, pose, or identity.
- Audio signals stem from underlying sources like speaker or background.
- We observe data x but wish to recover hidden causes z.

Goal: Learn a generative model that explains data via latent variables.

This is the idea behind **Variational Autoencoders (VAEs)**.

The Variational Autoencoder Framework

A VAE models the joint distribution:

$$p_\theta(x, z) = p_\theta(x \mid z)\, p(z)$$

- $x \in \mathbb{R}^d$ is the observed data.
- $z \in \mathbb{R}^n$ is the latent variable.
- $p(z)$ is usually a standard normal prior: $\mathcal{N}(0, I)$.
- $p_\theta(x \mid z)$ is a decoder (neural network).

The marginal distribution over data is:

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz$$

Training a VAE

Challenge: Computing $p_\theta(x)$ is often intractable due to the integral.

Solution: Use a variational approximation $q_\phi(z \mid x)$ and optimize the ELBO:

$$\begin{aligned}\log p_\theta(x) &\geq \mathcal{L}(q_\phi, \theta) := \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x,z)}{q_\phi(z \mid x)}\right] \\ &\geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] \; - \; KL(q_\phi(z|x)||p(z))\end{aligned}$$

Key Point: VAEs match the observed data distribution $p_\theta(x)$ to the data—but make no guarantees about the correctness of z.

Connection to ICA and Prior Work

Nonlinear ICA: Hyvärinen and Pajunen (1999) proved nonlinear ICA is not identifiable without extra structure.
VAEs: Locatello et al. (2019) showed empirically that disentanglement is impossible without inductive biases.
These results imply that:

Key Insight

Standard VAEs with unconditional priors cannot uniquely recover the latent factors that generate the data.

Motivation

- VAEs are generative models trained to approximately maximize the data log-likelihood via a variational lower bound (ELBO), so they can generate new samples.
- ICA focuses on recovering independent latent variables; it's not typically used as a flexible, high-quality generative model.

Can a VAE learn the joint distribution while retaining the generative capabilities of a VAE?

Deep Latent Variable Models: Formal Setup

We assume data are generated from latent variables through a true model:

$$p_{\theta^*}(x, z) = p_{\theta^*}(x \mid z)\, p_{\theta^*}(z)$$

True generative process:

$$z^{(i)} \sim p_{\theta^*}(z), \qquad x^{(i)} \sim p_{\theta^*}(x \mid z^{(i)})$$

We observe only the dataset:

$$\{x^{(1)}, \ldots, x^{(N)}\} \sim p_{\theta^*}(x)$$

Goal: recover the *sources* (true latent variables) or at least recover the true joint distribution.

Marginal Likelihood and VAEs

Given a parametric model:

$$p_\theta(x, z) = p_\theta(x \mid z)p(z) \tag{1}$$

The marginal over data is:

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, dz \tag{2}$$

Training (e.g., with ELBO) enforces:

$$p_\theta(x) \approx p_{\theta^*}(x) \tag{3}$$

Important: Matching $p(x)$ does not guarantee we recover the true z.

Parameters vs Neural Network Weights

The paper distinguishes two concepts:

- **Function-space parameters** θ represent the true generative mapping

$$(z \mapsto p_\theta(x \mid z))$$

- **Neural network weights** w are a particular implementation of θ via a network.

Two different weight vectors w and w' may implement the *same function* $p_\theta(x \mid z)$.
Therefore: Identifiability is defined in *function space*, not parameter space.

Unidentifiability: Rotation Example

Consider a VAE with prior $p(z) = \mathcal{N}(0, I)$. This prior is rotationally symmetric:

$$p_\theta(z) = p_\theta(Rz) \quad \forall R \in O(n)$$

Define a transformed decoder:

$$\tilde{p}_{\tilde{\theta}}(x \mid z) = p_\theta(x \mid R^{-1}z)$$

Then:

$$\tilde{p}_{\tilde{\theta}}(x) = \int p_\theta(x \mid R^{-1}z)\, p_\theta(z)\, dz = p_\theta(x)$$

Conclusion: Different $(\theta, \tilde{\theta})$ give the same distribution $\implies$ model is not identifiable.

Unconditional Priors

Theorem

Any latent variable model with an unconditional prior

$$p(z) \text{ does not depend on } u$$

is **not identifiable**.

Proof Overview:

- Gram-Schmidt-esque process to transform latent variables into linearly independent Gaussian R.V., $g(z)$
- Apply orthogonal transformation, M
- Invert transformation, $g^{-1}(z)$

Result: $z' = g^{-1}(Mg(z))$, with same distribution as original z.

What is Identifiability?

Identifiability: Can we uniquely recover the model from data?

Definition

A model is identifiable if for all (x, z): $\forall(\theta, \theta')$ such that

$$\int p_\theta(x, z)dz = \int p_{\theta'}(x, z)dz \quad \Rightarrow \quad p_\theta(x, z) = p_{\theta'}(x, z) \Rightarrow \theta = \theta'$$

In other words, matching marginal distributions implies we've recovered the true latent structure.

In VAEs: Even if $p_\theta(x)$ matches the data perfectly, the latent variables z can remain ambiguous.

Conditional Factorized Prior

- **Proposed identifiable model:** Factorized Univariate Exponential family distribution

$$p_{\lambda,\mathbf{T}}(\mathbf{z}|\mathbf{u}) = \prod_i \exp\left\{\langle \mathbf{T}_i(z_i), \lambda_i(\mathbf{u})\rangle - \psi_i(\lambda_i(\mathbf{u}))\right\} Q_i(z_i)$$

- Breaks symmetry of $p(z)$ across different values of u.
- Note that the natural parameters λ_i depends on the the observed variable $\mathbf{u}$.
- The dimension of the natural parameter vector λ_i is fixed at k for all i.
- With $\mathbf{x} = \mathbf{f}(\mathbf{z}) + \epsilon$, an injective but not necessarily linear mixing function plus a noise, we have

$$p_\theta(\mathbf{x}, \mathbf{z}|\mathbf{u}) = p_{\mathbf{f}}(\mathbf{x}|\mathbf{z}) p_{\lambda,\mathbf{T}}(\mathbf{z}|\mathbf{u}) \tag{5}$$

Equivalence Relation

Definition 2

Let $\sim$ be the equivalence relation on Θ defined as:

$$(f, T, \lambda) \sim (\tilde{f}, \tilde{T}, \tilde{\lambda}) \iff \exists A, c | T(f^{-1}(x)) = A\tilde{T}(\tilde{f}^{-1}(x)) + c, \forall c \in \mathcal{X}$$

where A is an $nk \times nk$ matrix and c is a vector. If A is invertible, denote by $\sim_A$.
If A is a block permutation matrix, denote by $\sim_p$.

Identifiability up to Equivalence Relation

Requirements for Identifiability

Theorem 1

1. The set $\{x \in X | \varphi_\epsilon(x) = 0\}$ has measure zero, where φ_ϵ is the noise characteristic function.
2. The mixing function f is injective.
3. The sufficient statistics $T_{i,j}$ are differentiable almost everywhere and are linearly independent on any subset of $\mathcal{X}$ of measure greater than zero.
4. There exist $nk + 1$ distinct points $u_0, \ldots, u_{nk}$ such that the matrix

$$L = (\lambda(u_1) - \lambda(u_0), \ldots, \lambda(u_{nk}) - \lambda(u_0))$$

is invertible.

Then the parameters (f, T, λ) are $\sim_A -$*identifiable*

Identifiability up to Equivalence Relation

Intuition of the Proof:

1. From the assumptions we can use a simple convolution trick to transform the observed data distributions into noiseless distributions.
2. Remove all terms that either a function of observations x or the auxiliary variables u. Using the distinct points with u_0 as a pivot.
3. Show the resulting linear transformation is invertible which proves identifiability.

Identifiability up to Equivalence Relation

Key Insight

Noiseless Case: $\tilde{f}$ transforms the observations into latents $\tilde{z} = \tilde{f}^{-1}(x)$ that are equal to the true generative latents $z = f^{-1}(x)$ up to linear invertible transformation and pointwise nonlinearities

With Noise: Same as the noiseless case from step 1 of proof.

Characterization of Linear Indeterminacy

Can be reduce equivalence up to permutation and signed scaling

Theorem 2 ($k \geq 2$)

Same assumptions as Thm 1 in addition:

1. The sufficient statistics $T_{i,j}$ are twice differentiable.
2. The mixing function f has all second order cross derivatives.

then the parameters (f, T, λ) are $\sim p - identifiable$

Theorem 3 ($k = 1$)

Same assumptions as Thm 1 in addition:

1. The sufficient statistics $T_{i,j}$ are not monotonic.
2. All partial derivatives of f are continuous.

then the parameters (f, T, λ) are $\sim p - identifiable$

When it can't be resolved

Proposition 1

Assume k=1, if:

1. $T_{i,1}(z_i) = z_i$ for all i
2. $Q_i(z_i) = 1$ or $Q_i(z_i) = \exp\{-z_i^2\}$ for all i

Then matrix A cannot reduced to a permutation matrix.

Example

The previous example of a rotation of a symmetric Gaussian distribution, with prior $p(z) = \mathcal{N}(0, I)$

$$p(z_i \mid u) = Q_i(z_i) \exp\{\lambda_i(u) z_i - \psi(\lambda_i(u))\}$$
$$Q_i(z_i) = e^{-z_i^2/2}$$
$$T_{i,1}(z_i) = z_i$$

Consistency of Estimation

These theorems also apply a consistency result on the VAE, if the variational distribution is a broad parametric family that includes the true posterior then

Theorem 4

Assume:

- The family of distributions $q_\phi(z|x, u)$ contains $p_{f,T,\lambda}(z|x, u)$
- We maximize $\mathcal{L}(\theta, \phi)$ with respect to both θ and ϕ

then in the limit of infinite data, the VAE learns the true parameters up the equivalence class.

Experiment 1: Synthetic data

- **Dataset** Non-stationary 2D Gaussian time-series sources $\mathbf{s}$, divided in M segments which labels $\{1, \ldots, M\}$ will be used as the conditioning variable $\mathbf{u}$. The true natural parameters of each segment are randomly and independently generated. Observations $\mathbf{x} = f(\mathbf{s}) + \epsilon$.
- **Model specification** The variational posterior (encoder) is of the form

$$q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{u}) = \mathcal{N}\left(\mathbf{z}|\mu(\mathbf{x}, \mathbf{u}; \phi), \mathbf{diag}\, \sigma^2(\mathbf{x}, \mathbf{u}; \phi)\right)$$

- **Performance** Compute the mean correlation coefficient (MCC) between original sources $\mathbf{s}$ and samples $\hat{\mathbf{z}}$ from the learned posterior q_ϕ

Experiment 1: Synthetic data

- **Results**

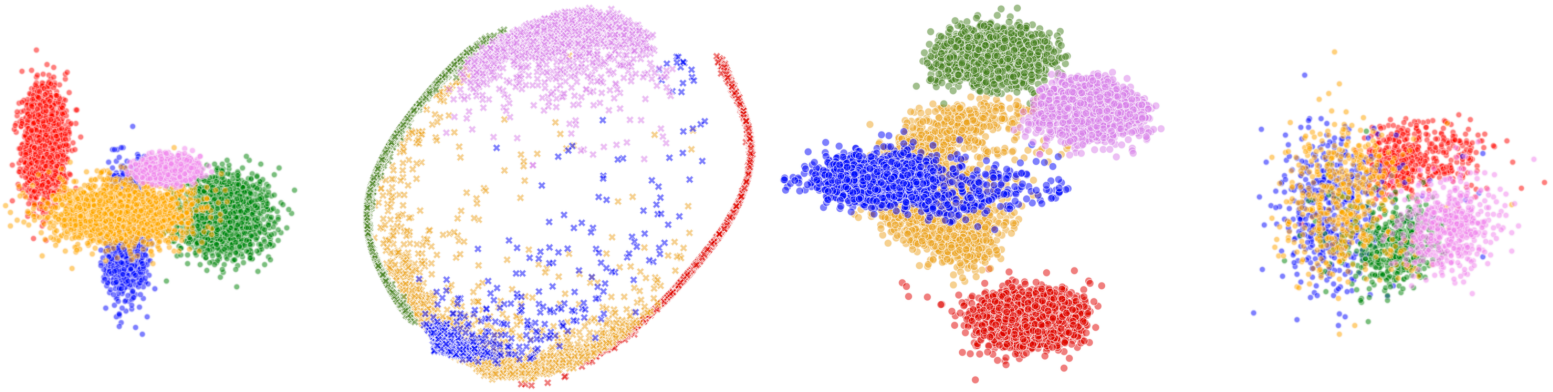
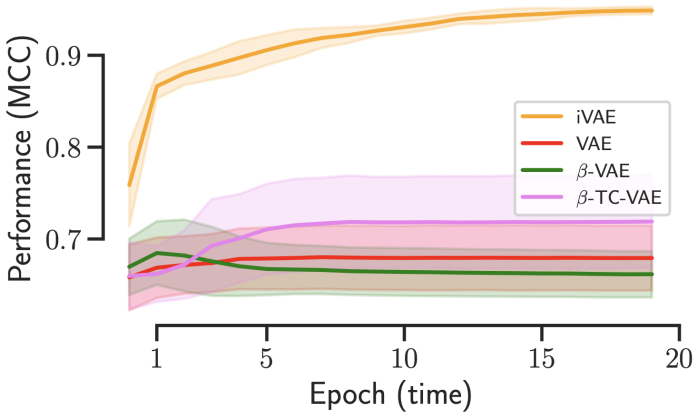


Figure: 2D example with 5 segments (colors). From left to right: (A) sources sample from true distribution $p_{\theta^*}(\mathbf{s}|\mathbf{u})$, (B) $\mathbf{x} = f(\mathbf{s})$, (C) sample from iVAE posterior $p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{u})$, (D) sample from vanilla VAE posterior $p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{u})$



Conclusion

- VAEs are great at approximating the data distribution; ICA methods are capable of learning the original latent code that generated the data up to a permutation, scale, and sign (identifiability).
- iVAE bridges these two goals by providing an identifiable generative model under certain conditions: (1) conditioning variable $\mathbf{u}$, (2) appropriate prior/likelihood design.
- iVAE can also be used for dimensionality reduction and for modeling noisy observations.
- The conditioning variable $\mathbf{u}$ in practice cannot be arbitrary; it must carry extra information about the latent space
  - Example, a class-conditional VAE used for image recognition.